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Management of Abnormalities of the External Genitalia in Boys

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Keywords

adolescent; buried penis; child; circumcision; hernia; hydrocele; micropenis; penis; prepuce; scrotum; testis torsion; testis; varicocele

Questions

1. In the male, which of the following stimulates the development of the external genitalia?
    a. Testosterone
    b. Human chorionic gonadotropin
    c. Dihydrotestosterone
    d. Luteinizing hormone (LH) and follicle-stimulating hormone (FSH)
    e. Maternal progesterone
2. What percentage of uncircumcised boys will have persistent phimosis by 17 years of age?
    a. Less than 1%
    b. 5%
    c. 10%
    d. 15%
    e. 20%

3. Circumcision should not be performed in neonates with which condition of the genitalia?
    a. Phimosis
    b. Undescended testis
    c. Inguinal hernia
    d. Penile curvature
    e. Testicular atrophy
4. What is the most common complication associated with circumcision?
    a. Trauma to the glans
    b. Bleeding
    c. Meatal stenosis
    d. Skin bridges
    e. Balanitis xerotica obliterans (BXO)
5. A 4-year-old boy presents with phimosis and BXO of the prepuce. What is the preferred treatment?
    a. Observation
    b. Topical corticosteroids
    c. Excision of BXO skin without circumcision
    d. Warm baths
    e. Circumcision
6. Penile agenesis is associated with all of the following malformations EXCEPT:
    a. cryptorchidism.
    b. vesicoureteral reflux.
    c. horseshoe kidney.
    d. ureteropelvic junction obstruction.
    e. renal agenesis.
7. The etiology of the buried penis includes all of the following EXCEPT:
    a. suprapubic fat pad.
    b. small penis.
    c. poor penopubic fixation of the penis.
    d. obesity.
    e. cicatrix scar after surgery.

8. A 9-month-old boy who was previously circumcised presents with a buried penis resulting from cicatricial scarring. What is the most appropriate initial treatment?
    a. Topical betamethasone and manual retraction
    b. Revision of circumcision
    c. Penopubic fixation of the penis
    d. Liposuction of the suprapubic fat pad
    e. Observation
9. What is the minimal normal stretched penile length of a full-term neonate?
    a. 1.2 cm
    b. 1.9 cm
    c. 2.5 cm
    d. 3.2 cm
    e. 4.5 cm
10. Which of the following statements is TRUE regarding a micropenis in a term male neonate?
    a. The testes are usually normal in size and not cryptorchid.
    b. It is best managed by gender reassignment.
    c. It has an abnormal ratio of the length of the penile shaft to circumference.
    d. It is unlikely to respond to testosterone stimulation until puberty.
    e. It is less than 1.9 cm in stretched length.
11. What is the most common cause of micropenis?
    a. Hypergonadotropic hypogonadism
    b. Hypogonadotropic hypogonadism
    c. Idiopathic
    d. Growth hormone deficiency
    e. Androgen insensitivity syndrome
12. Which of the following statements regarding penile masses is FALSE?
    a. The treatment of parameatal urethral cysts is complete excision of the cyst.
    b. The most common acquired cystic lesion of the penis is smegma under the unretractable prepuce.

c. Congenital penile nevi tend to be malignant.
d. The initial management of juvenile xanthogranulomas is expectant monitoring.
e. Epidermal inclusion cysts may form after penile surgery.

13. A 13-year-old African American boy with sickle cell disease has a 6-hour painful erection. The initial management should include all of the following EXCEPT:
    a. alkalization.
    b. hydration.
    c. intracavernous injections of β-adrenergic sympathomimetic agents.
    d. analgesia.
    e. transfusion to reduce hemoglobin S concentration.
14. Which of the following statements is TRUE regarding high-flow priapism?
    a. It is usually a drug-induced etiology.
    b. The aspirated blood is similar to venous blood on blood gas analysis.
    c. Color Doppler ultrasonography commonly demonstrates the fistula.
    d. Surgical intervention is the initial management.
    e. Sickle cell disease is the most common cause
15. Penoscrotal transposition is associated with all of the following anomalies EXCEPT:
    a. sex chromosome abnormalities.
    b. distal shaft hypospadias with chordee.
    c. sex chromosome abnormalities.
    d. Aarskog syndrome.
    e. caudal regression.
16. All of the following are associated with patency of the processus vaginalis EXCEPT:
    a. transverse testicular ectopia.
    b. epididymal anomalies.
    c. cryptorchidism.
    d. spermatic cord torsion.
    e. polyorchidism.

17. Abdominoscrotal hydrocele is reported to be associated with all of the following features EXCEPT:
    a. a closed processus vaginalis.
    b. epididymal anomalies.
    c. testicular dysmorphism.
    d. hydronephrosis.
    e. increased pressure within the tunica vaginalis.
18. Irreversible ischemic injury of the testicular parenchyma may begin as early as how many hours after torsion of the spermatic cord?
    a. 1
    b. 2
    c. 4
    d. 6
    e. 8
19. Which of the following is most specific in diagnosing spermatic cord torsion?
    a. High-riding testis
    b. Absence of the cremasteric reflex
    c. Transverse lie of the testis
    d. Spermatic cord twist on high-resolution Doppler ultrasonography
    e. Acute severe pain
20. After manual detorsion of the spermatic cord, which of the following is appropriate management?
    a. Color Doppler ultrasonography
    b. Radionuclide scan
    c. Doppler examination of the testis and spermatic cord
    d. Discharge from the hospital and arrangement for an office reevaluation in 1 week
    e. Immediate scrotal exploration
21. An adolescent is evaluated for a history of self-limited, intermittent episodes of severe unilateral scrotal pain. Physical examination findings are normal. What is the most appropriate course of action?
    a. Color Doppler ultrasonography
    b. Reassessment in 6 months
    c. Elective scrotal exploration

d. Radionuclide scrotal imaging
e. Immediate scrotal exploration

22. When the diagnosis of torsion of the appendix epididymis is made, which of the following is optimal management?
    a. Observation
    b. Color Doppler ultrasonography
    c. Radionuclide scrotal imaging
    d. Immediate scrotal exploration
    e. Cord block and manual detorsion
23. Which of the following is the most likely diagnosis in an infant with sterile urine and epididymitis?
    a. Unilateral renal agenesis
    b. Large prostatic utricle
    c. Urethral stricture disease
    d. Persistent vasoureteral fusion
    e. Radiographically normal urinary tract
24. What is the most appropriate course of action in an otherwise healthy neonate with perinatal testicular torsion?
    a. Surgical exploration of the affected testis
    b. Surgical exploration of the affected testis with contralateral scrotal orchidopexy
    c. Color Doppler ultrasonography of the scrotum
    d. Radionuclide testicular scan
    e. Observation
25. Most adolescent varicoceles evaluated by urologists are:
    a. painful.
    b. of cosmetic concern.
    c. asymptomatic.
    d. associated with an ipsilateral hydrocele.
    e. bilateral.
26. Significant testicular volume differential in cases of varicocele is defined as greater than:
    a. 5%.
    b. 5% to 10%.
    c. 10% to 15%.
    d. 15% to 20%

e. 25%.

27. Hydrocele formation after varicocele ligation is least likely to occur after which of the following procedures?
    a. Retroperitoneal ligation
    b. Subinguinal ligation
    c. Laparoscopic ligation
    d. Microscopic inguinal ligation
    e. Transvenous embolization
28. Which of the following is NOT a relative indication for elective varicocele repair?
    a. Pain
    b. Oligospermia
    c. Small testes
    d. Continuous spermatic venous reflux
    e. Testicular size discrepancy of greater than 20%

Answers

1. **c. Dihydrotestosterone.** Influence of dihydrotestosterone on the androgen receptors results in the differentiation of the genital tubercle, genital (labioscrotal) folds, and genital swelling at between 9 and 13 weeks of gestation into the male structures of the glans penis, penile shaft, and scrotum, respectively.
2. **a. Less than 1%.** Preputial retractability increases with age with 90% of uncircumcised boys 3 years of age with completely retractable prepuces; less than 1% by 17 years of age have phimosis.
3. **d. Penile curvature.** Circumcision should not be performed in neonates with other penile conditions that require surgical correction. These conditions include hypospadias, penile curvature, dorsal hood deformity, buried penis, and webbed penis.
4. **b. Bleeding.** The risk of complications after circumcision is 0.2% to 5%. The most common complication is bleeding, which occurs in 0.1% and is more common in older children.

5. **e. Circumcision.** Treatment of BXO includes medical and surgical management. The use of topical corticosteroids has had limited benefit to treat mild BXO of the prepuce with minimal scar formation. Circumcision is the preferred treatment.
6. **d. Ureteropelvic junction obstruction.** Penile agenesis (aphallia) results from failure of development of the genital tubercle. The disorder is rare and has an estimated incidence of 1 in 10 to 30 million births. The karyotype almost always is 46,XY, and the usual appearance is that of a well-developed scrotum with descended testes and an absent penile shaft. The anus is usually displaced anteriorly. Associated malformations are common and include cryptorchidism, vesicoureteral reflux, horseshoe kidney, renal agenesis, imperforate anus, and musculoskeletal and cardiopulmonary abnormalities.
7. **b. Small penis.** A buried penis can be classified into three categories based on etiology for the concealment: (1) poor penopubic fixation of the skin at the base of the penis; (2) obesity; and (3) a trapped penis from a cicatrix scar after penile surgery, typically a circumcision.
8. **a. Topical betamethasone and manual retraction.** Young children with secondary cicatrix after penile surgery can undergo forceful dilation of the cicatrix with a fine hemostat in the office after the application or injection of local anesthesia. Another option is the combination of topical betamethasone and manual retraction.
9. **b. 1.9 cm.** Stretched penile length is determined by measuring the penis from its attachment to the pubic symphysis to the tip of the glans. One must be careful to depress the suprapubic fat pad completely to obtain an accurate measurement, especially in an obese infant or child. In general, the penis of a full-term neonate should be at least 1.9 cm long, which is 2.5 SD below the mean of 4.0 cm.
10. **e. It is less than 1.9 cm in stretched length.** Micropenis is a normally formed penis that is at least 2.5 SD below the mean size in stretched length for age. The ratio of the length of the penile shaft to its circumference is usually normal, but occasionally the corpora cavernosa are severely hypoplastic. The testes are usually

small and frequently cryptorchid while the scrotum is usually fused and often diminutive. A stretched penile length less than 1.9 cm long is consistent with a micropenis.

11. **b. Hypogonadotropic hypogonadism.** The most common cause of micropenis is hypogonadotropic hypogonadism, which is the failure of the hypothalamus to produce an adequate amount of gonadotropin-releasing hormone (GnRH). This condition may result from hypothalamic dysfunction, which can occur in Prader-Willi syndrome, Kallmann syndrome (genital-olfactory dysplasia), Laurence-Moon-Biedl syndrome, and the CHARGE association.
12. **c. Congenital penile nevi tend to be malignant.** Congenital penile nevi tend to be superficial and benign. Congenital penile nevi are pigmented lesions that can form on the glans and penile shaft. They tend to be superficial and benign and should be excised.
13. **c. Intracavernous injections of β-adrenergic sympathomimetic agents.** The initial treatment of low-flow priapism resulting from sickle cell disease is conservative with hydration, oxygenation, alkalization, analgesia, and transfusion with the goal of reducing hemoglobin S concentration. Evacuation of blood and irrigation of the corpora cavernosa along with intracavernous injections of α-adrenergic sympathomimetic agents, such as phenylephrine or epinephrine solution, can be a concurrent therapy. Surgical intervention to allow corporeal drainage by shunt procedures is indicated if there is a lack of response to medical therapy.
14. **c. Color Doppler ultrasonography commonly demonstrates the fistula.** High-flow priapism is usually due to perineal trauma, such as a straddle injury. Corporeal irrigation is diagnostic and therapeutic. Typically, the aspirated blood is bright red and the aspirate is similar to arterial blood on blood gas analysis. Color Doppler ultrasonography often will demonstrate the fistula. The initial management is observation because spontaneous resolution may occur. Superselective embolization of cavernous and penile arteries is the next line of therapy. If not, angiographic embolization is indicated.
15. **b. Distal shaft hypospadias with chordee.** Frequently, penoscrotal transposition occurs in conjunction with perineal, scrotal, or

penoscrotal hypospadias with chordee. Penoscrotal transposition has also been associated with caudal regression, sex chromosome abnormalities, and Aarskog syndrome. As many as 75% of patients with complete penoscrotal transposition and a normal scrotum have a significant urinary tract abnormality, including renal agenesis and dysplasia, and other nongenitourinary anomalies.

16. **d. Spermatic cord torsion.** Risk of torsion is associated with abnormal development of the tunica vaginalis but not patency of the processus vaginalis.
17. **b. Epididymal anomalies.** The processus vaginalis is closed in cases of abdominoscrotal hydrocele, and an elongated dysmorphic testis, increased pressure within the tunica vaginalis, and hydronephrosis have all been reported.
18. **c. 4.** Irreversible ischemic injury to the testicular parenchyma may begin as soon as 4 hours after occlusion of the cord.
19. **d. Spermatic cord twist on high-resolution Doppler ultra sonography.** Spermatic cord twist on high-resolution Doppler imaging is the most specific finding, that is, the least likely to be false positive, in spermatic cord torsion.
20. **e. Immediate scrotal exploration.** It should be remembered that manual detorsion may not totally correct the full rotation that has occurred and that prompt surgical exploration is still indicated. Further, studies have shown that while torsion of the cord generally occurs in a medial inward fashion (requiring detorsion in an outward “open book” fashion), one-third of cases show a torsion in the opposite direction. Thus the usual detorsion method may actually increase the torsion in some cases. Attempted detorsion should never delay a patient from going to the operating room.
21. **c. Elective scrotal exploration.** If the suspicion is strong that episodes of intermittent torsion and spontaneous detorsion have occurred, the author’s experience has been that the finding of a bell-clapper deformity at exploration can be expected. Elective scrotal exploration should be performed, with scrotal fixation of both testes.
22. **a. Observation.** When the diagnosis of a torsed appendage is confirmed clinically or by imaging, nonoperative management will

allow most cases to resolve spontaneously.

23. **e. Radiographically normal urinary tracts.** The majority of infants with epididymitis have sterile urine and apparently radiographically normal urinary tracts.
24. **b. Surgical exploration of the affected testis with contralateral scrotal orchidopexy.** Clearly, if the cause of scrotal swelling appears to be related to an acute postnatal event, all efforts should be made to pursue prompt surgical intervention. If torsion is confirmed, contralateral scrotal exploration with testicular fixation should be performed.
25. **c. Asymptomatic.** Most adolescent varicoceles are asymptomatic.
26. **d. 15% to 20%.** In adults and adolescents, testicular size (volume) should be approximately equal bilaterally, with the normal differential not being more than 15% to 20% volume. Asynchronous testicular growth has been shown to occur normally throughout puberty.
27. **e. Transvenous embolization.** Hydrocele formation is related to failure to preserve lymphatic vessels associated with the spermatic cord and its vessels. Hydrocele formation seems most common after retroperitoneal ligation, especially when a mass ligation technique is used, and is least likely to occur after transvenous embolization.
28. **d. Continuous spermatic venous reflux.** Significant pain associated with varicocele (despite scrotal support), bilateral small testes (low total testicular volume), and oligospermia are reasonable indications to proceed with repair in an adolescent male. The standard indication is ipsilateral testicular volume loss, or hypotrophy, of at least 15% to 20%, although this should be documented on serial yearly testicular examinations, because variable growth of the testes may occur during puberty. Recent studies, however, have shown that differential testicular volume is a poor correlate of eventual total motile sperm count. Without attempted paternity history, the best indicator of testis health regarding fertility is a semen analysis. Continuous spermatic venous reflux may be documented on color Doppler imaging but is not a specific indication for surgery.

Chapter Review

1. Primary phimosis resolves spontaneously in the vast majority of boys; fewer than 1% of 17-year-old boys have a phimotic prepuce.
2. Indications to facilitate preputial retractility include persistent primary phimosis, secondary phimosis, balanitis, posthitis, balanitis xerotica obliterans (BXO), and urinary tract infection (UTI).
3. The American Academy of Pediatrics acknowledges the potential health benefits of the circumcised penis thus warranting access to circumcision for families choosing it and endorse third-party payment for circumcision of male newborns.
4. Local or topical anesthetics should be administered routinely for circumcisions performed in the neonatal period.
5. Circumcisions should be deferred in the presence of other penile abnormalities detected in the newborn period that may require surgery.
6. Health benefits of circumcision include reduction in rates of penile cancer, sexually transmitted diseases including HIV, UTIs, balanitis, and phimosis.
7. Topical application of corticosteroids can relax the phimotic ring in 70%–80% of cases.
8. The risk of complications from circumcision is 0.2%–5% with bleeding being the most common.
9. Circumcision revisions require the child to be under general anesthesia.
10. Meatal stenosis occurs only in the circumcised penis.
11. Topical corticosteroids fail to resolve penile adhesions.
12. BXO is a chronic infiltrative and scarring (cicatrix) skin condition.
13. Micropenis is a normally formed penis that measures at least 2.5 SD below the mean size in the stretched length for age (<1.9 cm in neonates).
14. Buried penis is not synonymous with micropenis as the penis is of normal stretched length.
15. The most common causes of micropenis include hypogonadotropic hypogonadism, hypergonadotropic hypogonadism.

16. Patient evaluation includes karyotype, determination of serum LH and FSH concentrations; measurement of testosterone levels before and after hCG stimulation; serum studies of anterior pituitary function; and MRI of the brain to assess hypothalamus, anterior pituitary, and midbrain.
17. In adulthood, most men with micropenis have male gender identity and satisfactory sexual function.
18. Priapism can be ischemic (veno-occlusive, low flow), nonischemic (arterial, high flow), and stuttering (intermittent).
19. The most common cause for priapism in children is sickle cell disease.
20. The initial treatment of low-flow priapism due to sickle cell disease is conservative with hydration, oxygenation, alkalization, and transfusion. Evacuation of blood and irrigation of the corpora cavernosa along with intracavernous injection of alpha-adrenergic sympathomimetic agents can be done concurrently or after failure of conservative measures.
21. High-flow priapism is usually due to perineal trauma, such as a straddle injury. Angiographic embolization should be performed if spontaneous resolution fails to occur.
22. Transient but potentially prolonged periods (less than 2 h) of erection, stuttering priapism, can be treated with pseudoephedrine given at bedtime.
23. The prevalence of varicoceles in adolescent and adult men are similar, approximately 15%.
24. Tall thin males are more commonly noted to have varicoceles, implicating a genetic predisposition.
25. Varicoceles are commonly identified on examination and least commonly identified due to pain.
26. Testicular volume is most accurately measured using ultrasound.
27. Testicular hypotrophy of 15%–20% may arise in the presence of an ipsilateral varicocele that commonly resolves after varicocele repair but can resolve spontaneously as well.
28. Trends toward poorer sperm quality may be limited to a subset of affected males with varicocele, but grade and postoperative testicular catch-up growth do not reliably predict ultimate semen quality.

29. Surgical success rates are similar with retroperitoneal (open or laparoscopic) or subinguinal approaches. Lymphatic-sparing techniques offer the lowest rates of hydrocele formation. The majority of hydroceles require surgical intervention.
30. Inguinal hernias develop in 1%–5% of children, more commonly in boys and premature infants. The propensity for the right side (3:1) is attributed to the later descent of the right testicle. Female gender, prematurity, age younger than 1 year, and history of cryptorchidism are risk factors for bilaterality.
31. An intermittent bulge helps to distinguish a reducible inguinal hernia and communicating hydrocele from a scrotal hydrocele or hydrocele of the spermatic cord. If a bulge is not elicited at the time of examination, photographs of the bulge taken at home are diagnostically reliable.
32. Inguinal hernias should be repaired soon after diagnosis while hydroceles can often resolve spontaneously and warrant a period of observation.
33. Surgical repair can be performed through an inguinal, scrotal, or laparoscopic approach. The inguinal approach carries the highest success rates at this time.
34. The causes of recurrent hernias (0.5%–1%) include failure to properly identify or ligate the sac during the original procedure; a tear in the sac, leaving a strip of peritoneum along the cord; damage to the floor; a weal internal inguinal ring; or an unrecognized direct hernia at the original exploration.
35. Transperitoneal diagnostic laparoscopy offers a rapid, direct, and accurate inspection of the contralateral internal inguinal ring.
36. Delay in presentation for acute scrotal pain may stem from patient embarrassment or denial, but also from lack of public awareness regarding torsion and the possible consequences.
37. In cases where a reliable thorough history is taken and targeted physical examination is performed, a reliable diagnosis should be made in the majority of cases and adjunctive ultrasound serves to confirm a diagnosis.
38. Ultrasound features suggestive of torsion of the spermatic cord include absent blood flow, a “snail sign” (cord torsion), and

heterogeneity of the parenchyma.

39. Testicular salvage rates from spermatic cord torsion are inversely related to duration and have some relationship to the degrees of torsion. Thus, in cases where torsion is highly suspected, emergent surgery should not be delayed by imaging studies.
40. Spermatic cord torsion occurring in the perinatal period is rarely salvageable when identified in the nursery and should be managed emergently when a change in physical examination has occurred.
41. Appendage torsion is the most common cause of acute scrotum in prepubertal children. A “blue dot sign,” discoloration at the upper pole of the testis, may be seen through stretched scrotal skin. This is a self-limiting process requiring supportive care and analgesia management.

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Hypospadias

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Keywords

2-stage repair; complications; hypospadias; reoperative hypospadias single-stage repair; single-stage repair; surgical technique

Questions

1. Hypospadias is typically made up of which of the following components?
    a. A high risk of associated anomalies outside of the genitourinary tract
    b. Ventral penile curvature with an orthotopic urethral meatus
    c. An intact prepuce
    d. An ectopic location to the urethral meatus below the center of the glans
    e. A bifid scrotum
2. The level of hypospadias is typically defined by what parameter?
    a. At the initial office visit by identifying the location of the urethral meatus
    b. By calculating the level of penile curvature in the office with a goniometer
    c. Based on the location of the meatus when inserting a feeding tube in the operating room

d. After penile degloving, taking into account both the location of the urethral meatus and the degree of penile curvature
e. In the operating room by determining the degree of separation of the glans wings

3. Which of the following is true about urethral development?
    a. The urethral groove forms between the urethral folds at an early stage in urethral development
    b. The urethral folds fuse between gestation weeks 8 to 16 to form the urethra
    c. Preputial skin folds fuse to complete the foreskin prior to fusion of the urethra
    d. a and b
    e. b and c
4. Which of the following support the disruption of androgen signaling as at least a partial cause of hypospadias?
    a. A shorter anogenital distance (AGD) and severity of hypospadias
    b. Increased rate of hypospadias development in boys conceived via in-vitro fertilization
    c. An increased risk of hypospadias development in monozygotic male twins compared to singleton live male births
    d. The concordance of rising rates of hypospadias, poor sperm quality, and cryptorchidism
    e. All of the above
5. An 8-month-old male presents for evaluation of subcoronal hypospadias. During physical examination, the left testicle is palpated in the groin but cannot be manipulated into the scrotum. The next step is to:
    a. reexamine in 6 months to allow for testicular descent.
    b. perform ultrasonography to rule out testicular retraction.
    c. obtain a karyotype.
    d. schedule hypospadias repair now, and orchiopexy in 6 months.
    e. schedule orchiopexy now, with hypospadias repair in 6 months.
6. At the beginning of a penoscrotal hypospadias repair, after degloving but prior to urethral tubularization, you pass a urethral

catheter but are unable to enter the bladder. The most likely cause is:

a. a false passage in the posterior urethra.
b. an enlarged utricle.
c. partial urethral duplication ending in a blind pouch.
d. an elevated bladder neck.
e. proximal urethral stricture.

7. A 6-month-old infant without other known medical problems is referred for proximal penile hypospadias. He is also found to have ventral penile curvature, a deep scrotal cleft, and penoscrotal transposition, but both testes are normally situated in the dependent scrotum. The next step is:
    a. proceed with surgery.
    b. obtain a karyotype.
    c. order a voiding cystourethrogram (VCUG) to rule out the presence of a utricle.
    d. obtain renal sonography to rule out a renal anomaly.
    e. delay surgery until 12 months of age to allow for interval penile growth.
8. Which of the following is true regarding penile innervation?
    a. The pattern of penile innervation does not have a significant impact on the choice of technique for correction of penile curvature.
    b. The nerve bundles remain as two well-defined structures on the dorsal aspect of the penis.
    c. In hypospadias surgery, a perineal nerve block may provide a better option than a caudal block in an 8-year-old boy.
    d. The pattern of innervation permits the safe use of dorsal plication techniques for orthoplasty, when the sutures are placed in the midline.
    e. b and d.
    f. b, c, and d.
9. Two years after a midshaft hypospadias repair a potty-trained 4-year-old boy presents with two febrile urinary tract infections (UTIs). Which of the following is the next step?
    a. Recommend that the boy sits to void

b. A VCUG
c. A daily cone enema
d. Cystoscopy under general anesthesia
e. Reassurance

10. A patient with penoscrotal hypospadias has 90 degrees of ventral curvature prior to making the initial incision. After the penis is fully degloved, artificial erection reveals residual curvature of 15 degrees. The best next step in correcting the curvature is:
    a. transection of the urethral plate.
    b. perform dermal grafting of the corpora cavernosa at the point of maximal curvature and complete with a 2-stage repair.
    c. proceed with urethroplasty.
    d. perform a midline dorsal plication.
    e. perform midline dorsal plication and ventral tunica vaginalis graft corporoplasty, then complete a single-stage urethroplasty repair.
11. A 6-month-old male is referred for hypospadias. On examination he is found to have a dorsally hooded prepuce, ventral penile curvature, a glanular meatus, and a normal scrotum. The parents should be informed that straightening the penile curvature most likely will require:
    a. skin degloving and ventral dartos dissection.
    b. multiple dorsal midline plications.
    c. ventral corporal grafting.
    d. transection of the urethra.
    e. single dorsal plication and ventral corporal graft.
12. A 6-month-old male is referred for hypospadias. On examination he is found to have a dorsally hooded prepuce, 90 degrees of ventral penile curvature, a coronal meatus, and significant penoscrotal webbing. The parents should be counseled as follows:
    a. Penile curvature will likely be corrected with skin degloving and ventral dartos dissection.
    b. The risk for hypospadias complication repair is 5%.
    c. Only one procedure should be required.
    d. A karyotype should be performed.
    e. A two-stage repair may be required.

13. A 6-week-old infant is evaluated after newborn circumcision. The primary care physician expresses concern for a “bad circumcision.” On examination the meatus is coronal and the glans wings are separated. The meatus itself appears elongated with a deep groove. The parents report no excessive bleeding after the circumcision. There is no redundant shaft skin following the circumcision except in the ventral midline near the meatus. The family should be informed that:
    a. these findings indicate urethral injury during circumcision.
    b. urethroplasty will best be performed using a ventral preputial skin flap.
    c. circumcision most likely has affected vascularity to the redundant ventral skin, and so urethroplasty will best be done using the skin as a graft.
    d. following circumcision, urethroplasty will require buccal graft from the lower lip.
    e. their infant has a hypospadias variant.
14. The parents of a 6-month-old male with subcoronal hypospadias request foreskin reconstruction rather than circumcision. The parents should be informed that:
    a. preoperative testosterone therapy to enlarge the foreskin is recommended.
    b. complication rates are significantly greater with prepucioplasty.
    c. they should not retract the foreskin in the first 6 weeks after surgery.
    d. a gentle compression dressing should be used to minimize preputial edema after surgery.
    e. the foreskin most likely will be needed for urethroplasty and so circumcision will likely be necessary.
15. Each of the following is thought to reduce the likelihood for fistula development after hypospadias surgery EXCEPT:
    a. subepithelial suturing of the neourethra.
    b. 2-layer closure of the neourethra.
    c. use of monofilament sutures for urethroplasty.
    d. approximation of the corpus spongiosum over the neourethra.

e. placement of a dartos flap over the neourethra.

16. A mother reports her 7-year-old child who had previously undergone a two-stage Byars flap hypospadias repair now has a prolonged urinary stream and significant dribbling from his meatus after voiding. All of the following are true EXCEPT:
    a. at the time of the repair local tissue can often be used to correct the problem.
    b. he will have erectile dysfunction as an adult.
    c. at the time of the repair the meatus should be calibrated to rule out a distal obstruction.
    d. the patient may note a bulging or ballooning of the ventral penile shaft when voiding.
    e. this complication can also occur in the setting of an island onlay flap repair.
17. An 8-year-old had distal hypospadias repair as an infant. Initially after surgery he was thought to have a normal urinary stream, although the parents only rarely observed urination because he used diapers. During the past year the parents think his stream has slowed and notice he seems to have to "push" to empty his bladder. Examination reveals a faint white discoloration around the meatus that seems to extend deeper into the urethra. The most effective course of action includes:
    a. intraoperative biopsy with frozen section.
    b. meatotomy for meatal stenosis.
    c. flip-flap reoperative urethroplasty.
    d. application of topical steroids for 6 weeks.
    e. excision of the distal urethra with two-stage buccal graft urethroplasty.
18. A 1-year-old boy had a subcoronal hypospadias repair 6 months ago but has a 2-mm fistula at the site of the original meatus. Distance from the fistula to the neomeatus is approximately 4 mm, and the glans wings still appear approximated. The meatus appears patent and the meatus is calibrated to 8Fr in the operating room. The best treatment of the fistula is:
    a. midline incision through the neomeatus to the fistula with reoperative distal urethroplasty.

b. rotational skin flap fistula closure.
c. primary fistula closure covered with a ventral dartos barrier flap.
d. inlay buccal graft urethroplasty.
e. insertion of urethral catheter to facilitate secondary closure.

19. A 10-year-old boy presents with midshaft hypospadias after seven prior operations on his penis. He appears to have ventral curvature greater than 30 degrees and has been circumcised. There is visible scarring between the meatus and the glans, and there appears to be significant penoscrotal webbing. The best surgical approach for him is:
    a. a redo tubularized incised plate (TIP) operation.
    b. flip-flap reoperative urethroplasty.
    c. island onlay flap reoperative urethroplasty.
    d. inlay buccal graft urethroplasty.
    e. two-stage buccal graft urethroplasty.
20. A 6-year-old boy who had a tubularized preputial flap hypospadias repair as an infant presents with a slow urinary stream and stranguria worsening over the past year. Physical examination is unremarkable, but the peak flow is 3 mL/s with a postvoid residual of 75 mL (estimated bladder capacity of 240 mL). At surgery cystoscopy shows a 5-mm stricture near the original meatus. This stricture is best corrected by:
    a. urethral dilation.
    b. direct vision internal urethrotomy (DVIU).
    c. DVIU with CIC for 3 months.
    d. inlay buccal urethroplasty.
    e. staged buccal graft reoperation.
21. A 9-year-old prepubertal boy has failed multiple prior operations for penoscrotal hypospadias. Examination shows a distal shaft meatus, persistent ventral curvature, and a flattened ventral glans. During planned two-stage buccal graft reoperation, it becomes apparent that the entire neourethra will have to be excised back to the penoscrotal junction. The best plan for first-stage grafting is to:
    a. use cheek buccal tissue on the penile shaft and maintain the current remnant prepuce within the glans.

b. use cheek buccal tissue to graft the entire defect.
c. use lower lip buccal tissue for the glans and cheek tissue on the penile shaft.
d. use lip buccal tissue on the penile shaft and cheek tissue within the glans.
e. use ventral penile shaft skin to graft the entire defect.

22. Preoperative androgen stimulation in patients with severe hypospadias has resulted in all of the following except:
    a. decreased penile size (compared to men without hypospadias) in adulthood.
    b. increase in penile size including length and glans width.
    c. hormonal levels that return to normal shortly after treatment is discontinued.
    d. nonappreciable long-lasting side effects of treatment.
    e. transient thin pubic hair.
23. Parents report that their 1-year-old boy seems to be having difficulty urinating 3 months after TIP repair for coronal hypospadias. They have observed the stream once or twice and thought it looked thin. On examination the glans looks normal, except that the meatus appears small. In the operating room you attempt to calibrate the meatus, and an 8-Fr sound will not pass. The most likely cause for this complication is:
    a. balanitis xerotica obliterans (BXO).
    b. ischemia of the neomeatus.
    c. postoperative edema of the meatus.
    d. suturing the urethral plate too far into the distal glans.
    e. prolonged urethral stent placement.
24. Parents report that their 1-year-old boy seems to be voiding without any problems 6 months after Thiersch-Duplay repair for coronal hypospadias, although they have not seen the actual urinary stream because he is still in diapers. On examination you observe that the meatus appears small. The next step is:
    a. calibrate the meatus.
    b. obtain VCUG.
    c. schedule examination under anesthesia and meatotomy.

d. recommend reoperative urethroplasty, using either a ventral flip-flap if possible, or inlay buccal grafting from the lip.
e. begin daily urethral dilations for 6 weeks.

25. If not repaired, an isolated glanular hypospadias with no penile curvature and a normal caliber to the meatus will result in which of the following long-term issues?
    a. Difficulty with micturition
    b. Inability to inseminate during sexual intercourse
    c. Clinically significant meatal stenosis
    d. Painful erections
    e. None of the above
26. A 19-month-old male presents for evaluation of scrotal hypospadias. The mother has noted that his pupils seem enlarged, and she is concerned he might have developmental delay. He is crying during examination, hindering inspection of his eyes. He has scrotal hypospadias with a deep scrotal cleft, but both testes are in the scrotum. His evaluation before surgery should include:
    a. renal sonogram.
    b. testicular sonogram.
    c. VCUG.
    d. assessment of testosterone/dihydrotestosterone (T/DHT) ratio.
    e. measurement of anti-Müllerian hormone (AMH).
27. A 12-year-old boy had hypospadias reoperation 1 year ago that included a tunica vaginalis barrier flap over the neourethra harvested from the right testicle. He reports no problems voiding, but with erection the penis is pulled to the right side. The next step is to:
    a. reassure him the tension on his penis will resolve at puberty.
    b. make a small scrotal incision and transect the tunica vaginalis flap.
    c. make a midline penile incision and excise the tunica vaginalis flap.
    d. create a tunica vaginalis flap from the left testis to evenly distribute the tension on the penis during erection.
    e. instruct the patient to pull the penis toward the left during erection to relax contracted tissues.

28. A 14-year-old undergoes a first-stage buccal graft reoperation that involves grafting along the entire penile shaft. The next morning he is found to have visible hematoma under the shaft skin. The next step is:
    a. return immediately to the operating room to evacuate the hematoma.
    b. apply a compression dressing over the penis and scrotum.
    c. check coagulation profiles for bleeding diathesis.
    d. observe with continued bed rest.
    e. evacuate the hematoma and regraft the penile shaft.
29. A 6-month-old boy has proximal hypospadias. In the operating room, there is 50 degrees of ventral penile curvature after penile degloving and division of the urethral plate. All of the following are viable repair options EXCEPT:
    a. corporal grafting with a dermal graft and 2-stage Byars flap repair.
    b. 2-stage STAG (staged tubularized autograft) repair with ventral corporotomies.
    c. single-stage repair with tunica vaginalis graft corporotomy followed by Duckett tube.
    d. single-stage repair with dorsal plication and island onlay urethroplasty.
    e. none of the above.
30. For hypospadias complication repairs, all of the following are true EXCEPT:
    a. a coronal fistula with a thin glans should be repaired with a redo urethroplasty and glansplasty.
    b. a small penoscrotal fistula identified 12 months after repair can be closed by placement of a urethral catheter and secondary closure.
    c. a urethral diverticulum can be closed using local tissue as a barrier layer.
    d. recurrent penile curvature should be assessed in the operating room to determine if penile skin scarring, corporal disproportion, or both are the cause.

e. glans dehiscence may require buccal mucosa graft inlay in order to achieve a successful repair.

31. Regarding familial hypospadias:
    a. only hypospadias associated with malformation syndromes has familial recurrence.
    b. hypospadias will occur only if the mother was taking birth control pills containing progesterone shortly before she conceived.
    c. hypospadias will occur only if the father also has hypospadias.
    d. it will occur, because hypospadias is a Y-linked disorder.
    e. the odds are greatest that another son would not have hypospadias.

Answers

1. **d. An ectopic location of the urethral meatus below the center of the glans.** The vast majority of boys with hypospadias do not have anomalies outside of the genitourinary tract or a bifid scrotum, the latter of which is typically seen in proximal hypospadias, which occurs 10% of the time. Some boys will have ventral curvature with an orthotopic urethral meatus, but they have chordee without hypospadias and can typically be corrected without repair of the urethra.
2. **d. After penile degloving, taking into account both the location of the urethral meatus and the degree of penile curvature.** The traditional method of diagnosis and risk stratification was based on the location of the urethral meatus at the time of diagnosis. Based on a number of different variants, the most accurate assessment of the degree of hypospadias is based upon the location of the urethral meatus after penile degloving combined with an assessment of the degree of penile curvature. Taking note of glans separation, although part of the assessment, is present in the majority of boys with hypospadias, and is not thought to correlate with the degree of hypospadias.

3. **d. a and b.** The urethral groove forms between the urethral folds and ultimately becomes the urethra. Preputial folds fuse as the urethra closes, in a dorsal to ventral fashion, hence the typical physical characteristics seen in hypospadias.
4. **e. All of the above.** A shorter AGD is thought to be secondary to exposure to environmental disrupting chemicals, currently theorized to inhibit the androgen cascade and penile development, resulting in hypospadias. The increased rate of hypospadias in twins and those boys conceived via in-vitro fertilization is thought, in part, to be secondary to hormone disruption. The testicular dysgenesis syndrome (TDS) includes several male reproductive developmental disorders, including poor sperm quality and rising rates of cryptorchidism and hypospadias, thought to be secondary to fetal exposure and resultant disruption of the normal androgen cascade during development, resulting in a life-long effect on these boys.
5. **c. Obtain a karyotype.** The combination of a penile anomaly with undescended testis may indicate a disorder of sex development. Although this is more commonly found with proximal hypospadias and a nonpalpable gonad, it is still advised to obtain a karyotype in any child with hypospadias and cryptorchidism.
6. **b. An enlarged utricle.** The most common reason for difficulty with catheter placement during hypospadias repair is an enlarged utricle, which occurs more frequently in boys with more severe hypospadias. Urethral duplication is rare in boys with hypospadias and a high bladder neck is encountered often in boys with posterior urethral valves, not hypospadias.
7. **a. Proceed with surgery.** Neither karyotyping nor urinary tract imaging is indicated for isolated hypospadias, even in patients with proximal defects. Associated renal anomalies are rare in nonsyndromic hypospadias. Karyotyping can be considered and offered to families with proximal hypospadias but is only strongly recommended in the setting of at least one undescended testicle. Delaying surgery will not significantly increase penile growth. If a utricle is encountered at the time of surgery maneuvers are taken to

successfully place a catheter and removal at the time of primary hypospadias repair is not recommended.

8. **f. b, c, and d.** The pattern of penile nerve innervation plays a critical role in management of penile curvature and must be completely understood prior to correction. The nerve bundles are present at the 11 and 1 o'clock positions on the dorsal penile shaft and can be avoided with midline suture placement. Perineal nerve block can be attempted in older, ambulatory patients who would not otherwise tolerate the temporary lower extremity paralysis after caudal blockade.
9. **b. A VCUG.** Recurrent UTIs after a hypospadias repair are a relatively rare event. Anatomic concerns including distal stricture, new onset vesicoureteral reflux, or incomplete voiding must be explored. Constipation may be contributing to the UTIs, but a cone enema is not recommended at this point. If an anomaly is seen on VCUG a cystoscopy should be considered. Sitting to void will not help this issue.
10. **d. Perform a midline dorsal plication. Persistent penile curvature of less than 30 degrees after penile degloving can be straightened by a single dorsal plication without clinically apparent shortening of the penis.** Transection of the urethral plate and/or ventral corporal lengthening procedures are reserved for cases with curvature greater than 30 degrees after degloving and dartos dissection. In a boy with 15 degrees of ventral curvature, correction is recommended and proceeding with urethroplasty significantly increases the chance that an additional operation to correct curvature will be necessary in the future.
11. **a. Skin degloving and ventral dartos dissection.** Most often penile curvature noted in so-called chordee without hypospadias is due to shortened ventral skin and dartos and so corrects as the penis is degloved and ventral dartos dissected. Multiple plications should be avoided, ventral corporal grafting is reserved for curvature greater than 30 degrees after degloving and ventral dartos dissection, and urethral transection for a shortened urethra is only rarely indicated in this condition.

12. **e. A two-stage repair may be required.** Although difficult to fully assess in the office setting, a patient with significant curvature and penoscrotal webbing may be suggestive of a severely hypoplastic urethra with significant penile curvature that may require a 2-stage repair. Parent counseling at the time of the initial visit is a key component of setting expectations for surgical reconstruction. A patient with severe ventral curvature is unlikely to have a complication rate of 5% and most series suggest a total complication rate closer to 10% even for distal hypospadias. A karyotype is not necessary in isolated hypospadias.
13. **e. Their infant has a hypospadias variant. The patient has megameatus intact prepuce (MIP) hypospadias variant, and urethroplasty is typically successful via tubularization of local skin flaps.** Preputial skin flaps or buccal grafts are typically not necessary. This is not secondary to a circumcision injury.
14. **c. They should not retract the foreskin in the first 6 weeks after surgery.** Disruption of the reconstructed prepuce may result from attempts to retract the foreskin early after surgery, before edema subsides and the wound heals. Otherwise complication rates are similar between patients with distal hypospadias undergoing foreskin reconstruction versus circumcision. Dressings do not significantly affect postoperative edema. When a sufficient amount of redundant shaft skin is present prepucioplasty can be performed, but the repair should not be compromised in order to achieve this. Tunica vaginalis can be considered for secondary layer closure in these patients.
15. **c. Use of monofilament sutures for urethroplasty.** No study demonstrates outcomes are influenced by suture type, but technical modifications such as subepithelial suturing of the neourethra and additional layers of vascularized tissue are beneficial.
16. **b. He will have erectile dysfunction as an adult.** The history of a 2-stage Byars flap and a prolonged voiding time with significant dribbling suggests a urethral diverticulum. It can be repaired using the redundant tissue present within the diverticulum itself, as some of the tissue can be de-epithelialized and used for additional layers of vascularized closure. They are thought to occur in preputial

onlay flaps as well as 2-stage Byars flap repairs due to the dispensability of the preputial skin.

17. **e. Excision of the distal urethra with two-stage buccal graft urethroplasty.** The history and physical findings suggest BXO. A biopsy in the operating room (OR) can be performed at the time of the procedure, but this is not a definitive process. Office biopsy is not recommended. **Meatotomy or skin flap repair of meatal stenosis most often fails in the presence of BXO, as residual BXO will progress into the reconstructed tissue. Given the extension of the suspected BXO into the urethra, a cystoscopy to determine proximal extent with subsequent excision of all tissues affected by BXO with staged buccal mucosa reconstruction is standard of care.** Topical steroids may be initially effective, but stenosis recurs when therapy ends, and BXO may extend proximally along the urethra to a level not reached by topical applications. In spite of replacement with buccal mucosa, these patients will still be at risk for recurrence and must be monitored closely.
18. **c. Primary fistula closure covered with a ventral dartos barrier flap.** A fistula with good glans approximation usually can be primarily corrected with fistula closure followed by coverage with additional vascularized layers, in this case a Dartos barrier flap. In contrast, when the skin quality is thin and poor with separation of the glans wings in addition to a short distance between the neomeatus and the fistula, this is not recommended. Instead a reoperative distal urethroplasty/glansplasty should be performed, otherwise a high failure rate will follow. A rotational skin flap is not a good option in this case because it is necessary to dissect under the corona to completely free the fistula tract and advance a barrier flap. Although meatal stenosis has to be considered in any case with a fistula, appropriate therapy when it is present is reoperation, not dilations. Prolonged catheter insertion will not lead to fistula closure.
19. **e. Two-stage buccal graft urethroplasty. Visible scarring is a relative contraindication to TIP or flip-flap reoperations, as there is likely significant compromise of the blood supply to**

these tissues, which will result in a high failure rate. The penoscrotal webbing and curvature suggests that there is a lack of shaft skin, particularly on the ventral portion of the penis, which will limit the ability to perform an onlay procedure The best plan is to excise scarred tissues and perform a staged buccal graft urethroplasty. The penile curvature must be assessed at the time of the repair to ensure that there is no residual curvature after the scar tissue is excised.

20. **d. Inlay buccal urethroplasty. Distal urethral narrowing, particularly within the glans in which there is lack of normal expansion of the urethra, can lead to an elevated postvoid residual and slowed urinary stream. Direct vision internal urethrotomy (DVIU) has a long-term success less than 10% after TIP repairs, even when used in conjunction with clean intermittent catheterization (CIC).** The best treatment in this case is inlay grafting into the stricture and either a buccal mucosa graft or a piece of redundant dorsal shaft skin or inner preputial skin can be used for a dorsal inlay graft repair. If an inlay graft repair fails, a staged buccal graft repair would be required.
21. **c. Use lower lip buccal tissue for the glans and cheek tissue on the penile shaft. It is best to use lip tissue within the glans, because it is thinner than cheek tissue and results in less complications after glansplasty.** Given the narrow and flat appearance of the glans and a history of multiple failed repairs, the skin with the glans is best excised to restore a deep groove that ultimately will result in a vertical meatus at the second stage. After multiple failed operations, redundant shaft skin sufficient for grafting and subsequent urethroplasty is not available.
22. **a. Decreased penile size (compared to men without hypospadias) in adulthood.** Preoperative androgen stimulation in prepubertal boys can transiently increase both the penile length and glans width. It is thought that a small glans width increases urethroplasty and glansplasty complications, therefore increasing the width of the glans at the time of closure is of some theoretical benefit. Several conflicting studies argue both sides of its use. Its side effects including increased frequency of erections, thin pubic

hair, and increased aggression, all of which are transient after it is discontinued.

23. **d. Suturing the urethral plate too far into the distal glans. The most common cause of meatal stenosis after TIP is suturing the urethral plate too far distally.** The contour of the glans is such that this will narrow the distal portion of the meatus, causing stenosis and/or obstruction. BXO is unlikely early after surgery and often presents with white discoloration at the meatus. Ischemia is possible but unlikely, given the reliable vascularity of the urethral plate and glans tissues. Edema after surgery may slow the urinary stream, but at 3 months it should be resolved enough to not cause passage of a sound. Prolonged urethral stent placement does not cause obstruction.
24. **a. Calibrate the meatus.** The meatus may appear small after Thiersch-Duplay urethroplasty without indicating meatal stenosis. Passage of an 8- or 10-Fr sound would suffice to determine if the meatus is stenotic or not, although this is extremely difficult in the office setting. Passage of a sound without anesthesia is only possible in a small number of patients, and even insertion of an 8-Fr feeding tube may be difficult in this population. A small-appearing meatus in an asymptomatic (UTI, stranguria, prolonged urinary stream) patient should not prompt immediate concern that examination under anesthesia or reoperation is needed. If symptoms are present an examination under anesthesia is recommended. If repair is indicated, the intervention of choice would depend upon the length of narrowing, the quality of the glans closure, and the assessment of the remainder of the urethroplasty.
25. **e. None of the above.** In isolated hypospadias without any evidence of narrowing of the meatus, a glanular location, and no penile curvature, it is unlikely that significant morbidity would result for the patient. If penile curvature is noted, however, even with a mildly displaced meatus, surgical intervention would be warranted due to the potential for long-term sequelae.
26. **a. Renal sonogram.** The history suggests WAGR (Wilms tumor, Aniridia, Genital abnormalities, mental Retardation) syndrome, and

so renal sonography is recommended because of the association with Wilms tumor.

27. **b. Make a small scrotal incision and transect the tunica vaginalis flap. Failure to dissect the tunica vaginalis flap to the external ring can result in traction on the penis during erection, deviating it toward the base of the flap.** The best therapy is to make a small scrotal incision over the flap and transect it. As this has occurred several years after his initial repair, this should not compromise the blood supply to the urethroplasty.
28. **d. Observe with continued bed rest. Hematoma formation under buccal mucosa graft can compromise the take rate of the graft. Technical modifications including graft quilting and the tie-over compression dressing prevent blood from accumulating under the graft. Hematoma under adjacent penile skin and within the scrotum as described here will resolve with minimal effect upon the graft itself.** It is unlikely that a boy requiring a staged buccal graft repair has an undiagnosed coagulopathy following prior penile operations.
29. **d. Single-stage repair with dorsal plication and island onlay urethroplasty.** In a boy with severe ventral curvature after penile degloving and division of the urethral plate (50 degrees in this case), a two-stage repair should be performed. A corporal lengthening procedure should be performed, and dermal graft, corporotomies, and tunica vaginalis flap are all viable options. Dorsal plication would put the boy at risk for recurrent penile curvature long term.
30. **b. A small penoscrotal fistula identified 12 months after repair can be closed by placement of a urethral catheter and secondary closure.** Fistulas rarely, if ever, close with catheter insertion. A coronal fistula with thin glans tissue will likely need a more aggressive repair. A urethral diverticulum can be de-epithelialized to add additional layers of closure. Recurrent curvature should be aggressively diagnosed and managed as there is a high rate of failure with plication alone.
31. **e. The odds are greatest that another son would not have hypospadias.** Sporadic and syndromic hypospadias can have

familial recurrence, and the risk of a boy with a brother with hypospadias having hypospadias himself is as high as 15%. This rate would be higher in a twin pairing. Although the likelihood a sibling will have hypospadias is increased, the overall risk remains small.

Chapter Review

1. Dihydrotestosterone at the 8- to 12-week gestational phase is a key mediator in the proper development of the penis.
2. There is an increased risk of hypospadias in births resulting from assisted reproduction and monozygotic twins.
3. The etiology of hypospadias is multifactorial, but placental disruption and androgen disruption, in part via environmental exposures, are potential culprits in what is thought to be an increasing rate of hypospadias development. Anogenital distance is a reflection of disruption of this cascade.
4. Parental discussion about the risk of hypospadias repair should include the risks of early anesthesia exposure versus the benefits of early surgery in infancy.
5. Preoperative androgen stimulation can transiently increase the penile length and size for surgery. Its impact on complication development is controversial.
6. Physical examination in a patient with hypospadias may reveal a ventral deficient prepuce, downward glans tilt, deviation of the median penile raphe, ventral curvature, scrotal encroachment onto the penile shaft, scrotal cleft, and penile scrotal transposition.
7. Eighty percent of urethroplasty complications occur within 1 year after surgery. Delayed complications, such as urethral stricture or penile curvature, can present in a delayed fashion. Thus boys, particularly those with proximal hypospadias, should be followed into puberty.
8. Risk factors for complications following urethroplasty include proximal meatus, reoperation, glans width less than 14 mm, and lack of a barrier flap over the neourethra.

9. Long-term follow-up of patients who have had a hypospadias repair indicates that they are more likely to have ejaculatory problems, are less satisfied with sexual function, and are more likely to be dissatisfied with the appearance of their penis than controls. Patient-reported outcomes are an underrepresented component of follow-up and must be a focus of future studies.
10. The combination of a penile anomaly with undescended testis may indicate a difference of sex development (DSD). Although more commonly found with proximal hypospadias and a nonpalpable gonad, it is still advised to obtain a karyotype in any child with hypospadias and cryptorchidism.
11. Curvature less than 30 degrees can be straightened by a single dorsal plication without clinically apparent shortening of the penis. Objective measurement of penile curvature, as identified by artificial erection after penile degloving, is an important part of the surgical procedure. Transection of the urethral plate and/or ventral corporal grafting are reserved for cases with curvature greater than 30 degrees after degloving and dartos dissection.
12. Meatotomy or skin flap repair of meatal stenosis most often fail in the presence of BXO, whereas excision of all tissues affected by BXO with staged buccal grafting is considered most likely to succeed without recurrent stenosis. These patients will have up to a 40% failure rate; therefore they must be closely monitored.
13. For buccal mucosal reconstruction, lip tissue placement is preferable within the glans, because it is thinner than cheek tissue and results in a lower complication rate.
14. Urethrocutaneous fistula is the most common complication after hypospadias repair. If there is supple tissue present, a primary closure can be attempted, but if thin glans tissue is present or located at the corona or distal, then redo urethroplasty and glans closure should be performed.

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Etiology, Diagnosis, and Management of Undescended Testis

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Keywords

cryptorchidism; gubernaculum; orchidopexy; subfertility; testicular descent; testicular germ cell tumor

Questions

1. What is the master gene responsible for male sexual differentiation?
    a. RSPO1
    b. SOX9
    c. WT1
    d. SRY
    e. WNT4
2. During male reproductive tract development, androgens mediate the differentiation of all of the following structures EXCEPT:
    a. seminal vesicles.
    b. ureter.
    c. epididymis.
    d. vas deferens.
    e. ejaculatory ducts.

3. Which of the following does not play a direct role in testicular descent?
    a. Testis
    b. Epididymis
    c. Genitofemoral nerve
    d. Gubernaculum
    e. Processus vaginalis
4. Peak levels of testosterone and insulin-like 3 occur in the male fetus at approximately what gestational week?
    a. 5
    b. 8
    c. 10
    d. 15
    e. 20
5. Cryptorchidism increases the risk of all of the following EXCEPT:
    a. spermatic cord torsion.
    b. clinical hernia.
    c. reactive hydrocele.
    d. infertility.
    e. testicular malignancy.
6. The risk of cryptorchidism is higher in all of the following syndromes EXCEPT:
    a. cerebral palsy.
    b. cystic fibrosis.
    c. arthrogryposis.
    d. prune belly syndrome.
    e. posterior urethral valves.
7. All of the following are associated with patency of the processus vaginalis EXCEPT:
    a. transverse testicular ectopia.
    b. epididymal anomalies.
    c. cryptorchidism.
    d. spermatic cord torsion.
    e. polyorchidism.
8. Histologic findings in cryptorchid testes may include all of the following EXCEPT:

a. carcinoma in situ.
b. absence of Ad spermatogonia.
c. early disappearance of gonocytes.
d. failure of Sertoli cell maturation.
e. reduced germ cell counts.

9. What percentage of undescended testes are nonpalpable at presentation?
   a. 1%
   b. 3%
   c. 10%
   d. 20%
   e. 30%
10. During laparoscopy, spermatic cord structures exiting an open internal ring ipsilateral to a nonpalpable testis imply:
    a. vanishing testis, inguinal exploration unnecessary.
    b. vanishing testis, inguinal exploration necessary.
    c. intracanalicular atrophic testis, inguinal exploration unnecessary.
    d. intracanalicular testis, inguinal exploration necessary.
    e. further exploration unnecessary if contralateral testicular hypertrophy is present.
11. Advantages of laparoscopic management of an intraabdominal testis include all of the following EXCEPT:
    a. more accurately assesses the presence or absence, viability, and anatomy of the testis compared with radiographic imaging.
    b. reduces the risk of damage to the testis.
    c. enhances surgical exposure, lighting, and magnification.
    d. allows a greater degree of proximal dissection of the spermatic vessels.
    e. allows diagnosis of associated Müllerian ductal abnormalities if present.
12. Which statement is FALSE regarding Fowler-Stephens orchidopexy?
    a. It is less commonly associated with testicular atrophy than laparoscopic primary orchidopexy.

b. It has a lower success rate in patients who have undergone previous inguinal surgery.
c. Blood supply is based on the deferential artery and collateral peritoneal vessels.
d. It should be performed at a similar age as a standard inguinal orchidopexy.
e. It should be considered if the testis is not near the internal ring.

13. Which of the following is least consistent with a diagnosis of vanishing testis?
    a. Patent processus vaginalis
    b. Contralateral testicular hypertrophy
    c. Palpable nubbin in scrotum
    d. Increased serum follicle-stimulating hormone
    e. Micropenis
14. Lower than expected testicular volume has been associated with all of the following EXCEPT:
    a. cryptorchid testes that have descended spontaneously.
    b. solitary testes in boys with a vanishing testis.
    c. Fowler-Stephens orchidopexy.
    d. increased serum FSH.
    e. surgery for congenital cryptorchidism at 3 years as compared to 9 months of age.
15. Regarding epididymal anatomy, which of the following is the most common finding in boys undergoing orchidopexy for acquired cryptorchidism?
    a. Detachment of the cauda epididymis
    b. Detachment of the caput epididymis
    c. Looped epididymis
    d. Long looping epididymis/vas
    e. Normal anatomy
16. All of the following factors may influence the reliability of studies of the efficacy of hormone therapy for cryptorchidism EXCEPT:
    a. treatment of boys with retractile testes.
    b. initial position of the testis.
    c. vanishing testis syndrome.

d. randomization protocol.
e. all of the above.

17. A healthy full-term newborn boy presents with a bilateral nonpalpable testes. The next step in management should include:
    a. karyotype analysis.
    b. GnRH stimulation test.
    c. abdominal ultrasound.
    d. circumcision.
    e. discharge home with MRI with gadolinium in four weeks.
18. Of the following, which is the LEAST reliable test in confirming the diagnosis of bilateral anorchia?
    a. No change in serum testosterone following hCG stimulation
    b. FSH level >2 IU/L at 1 year of age
    c. Laparoscopy
    d. Undetectable serum inhibin B
    e. Undetectable serum anti-Müllerian hormone (AMH)
19. Which of the following is LEAST useful to the provider in determining the diagnosis of retractile vs. undescended testes?
    a. Observation of testicular position with abduction of the patient's legs
    b. History of prior testicular position provided by the primary care physician (PCP)
    c. Failure of the testis to remain stable in the scrotum with sustained traction on the cord
    d. Warm room and hands
    e. Small ipsilateral testis
20. Which of the following is true regarding spontaneous descent of cryptorchid testes?
    a. Spontaneous descent is independent of testicular position.
    b. Reascent occurs in 40% of patients.
    c. Early descent is more likely in premature boys.
    d. Spontaneous descent is unlikely if the ipsilateral hemiscrotum is small.
    e. The majority of testes that descend spontaneously do so in the first few months of life.

21. The risk of developing a testicular tumor in males with a history of cryptorchidism is:
    a. 2 to 5 times the risk in normal boys.
    b. minimal in boys who undergo orchidopexy in infancy.
    c. determined by placental alkaline phosphatase (PLAP) staining in prepubertal testes.
    d. similar in the contralateral descended testis.
    e. increasing over time.
22. Each of the following is more common in association with cryptorchid testes, EXCEPT:
    a. atrophy.
    b. microlithiasis.
    c. mature teratoma.
    d. ectasia of the rete testis.
    e. intratesticular varicocele.
23. The following are possible locations of an ectopic testis, EXCEPT:
    a. peripenile.
    b. perirenal.
    c. perivesical.
    d. perianal.
    e. femoral.
24. Levels of all the following hormones peak after birth and remain low during childhood EXCEPT:
    a. LH.
    b. FSH.
    c. AMH.
    d. Inhibin B.
    e. testosterone.
25. When in fetal development does the testicle pass into the inguinal canal?
    a. 5 to 7 weeks gestation
    b. 10 to 14 weeks gestation
    c. 20 to 28 weeks gestation
    d. 30 to 34 weeks gestation
    e. None of the above
26. How commonly does cryptorchidism occur in full-term males?

a. <1%
b. 1% to 4%
c. 5% to 10%
d. 15%
e. None of the above

27. A 1-month-old full-term male presents with a unilateral nonpalpable testicle. The next step after a confirmatory exam is:
    a. ultrasound to identify the position of the testicle.
    b. MRI with gadolinium.
    c. hormonal therapy to induce descent.
    d. surgical intervention.
    e. observation for spontaneous descent until 6 months of age.

Answers

1. **d. SRY.** The SRY gene appears to be primarily responsible for male sexual differentiation through complex interactions involving both activation and repression of other male-specific genes.
2. b**. Ureter.** Androgens (testosterone, dihydrotestosterone) mediate the differentiation of the paired Wolffian ducts into the seminal vesicles, epididymis, vas deferens, and ejaculatory ducts.
3. **b. Epididymis.** Changes in the gubernaculum and processus vaginalis and their innervation by the genitofemoral nerve, and hormone secretion by the testis are all important in the process of testicular descent.
4. **d. 15.** Testosterone production peaks at 14 to 16 weeks and INSL3 peaks at 15 to 17 weeks.
5. **c. Reactive hydrocele.** All of the others are possible complications of cryptorchidism.
6. **b. Cystic fibrosis.** All the other syndromes are associated with a higher risk of cryptorchidism; mutations of the cystic fibrosis gene are associated with congenital absence of the vas deferens.
7. d. **Spermatic cord torsion.** Risk of torsion is associated with abnormal development of the tunica vaginalis but not patency of the processus vaginalis.

8. **c. Early disappearance of gonocytes.** Histological abnormalities that may be present in cryptorchid testes include delayed disappearance of gonocytes, reduced numbers of adult dark (Ad) spermatogonia, reduced number of germ cells per testicular tubule, and carcinoma in situ (CIS).
9. **d. 20%.** Approximately 20% of undescended testes are nonpalpable at presentation.
10. **d. Intracanalicular testis, inguinal exploration necessary.** While atretic spermatic vessels seen exiting the internal ring may be associated with a distal vanishing testis, the appearance of the spermatic vessels during laparoscopy is subjective and therefore exploration (inguinal or laparoscopic) is needed to rule out an intracanalicular viable or atrophic testis. Further exploration is unnecessary if blind-ending intraabdominal spermatic vessels are found. Hypertrophy of a normally descended contralateral testis is suggestive of monorchism.
11. **b. Reduces the risk of damage to the testis.** Laparoscopic orchidopexy still carries the risks of bleeding, infection, and damage to the testis and cord in addition to damage to intraabdominal structures such as bowel and the great vessels.
12. **a. It is less commonly associated with testicular atrophy than laparoscopic primary orchidopexy.** Fowler-Stephens orchidopexy, either one-or two-stage, has a higher reported testicular atrophy rate compared with laparoscopic orchidopexy. The other statements are true.
13. a. **Patent processus vaginalis.** Contralateral testicular hypertrophy and a palpable scrotal nubbin may present in boys with unilateral vanishing testis and increased serum FSH and micropenis may be seen in boys with bilateral vanishing testes. The processus vaginalis is closed in most cases of vanishing testis.
14. b. **Solitary testes in boys with a vanishing testis.** In vanishing testis syndrome, the contralateral testis may be larger than expected for age.
15. e. **Normal anatomy.** An abnormal epididymis was reported ipsilateral to 11% to 31% of acquired undescended testes at surgery.

16. **e. All of the above** are confounding factors that may affect the reliability of studies of the efficacy of hormone therapy for cryptorchidism.
17. **a. Karyotype analysis.** AUA Guidelines for cryptorchidism state that providers must immediately consult an appropriate specialist for all phenotypic male newborns with bilateral, nonpalpable testes for evaluation of a possible difference of sex development (DSD). A newborn with a male phallus and bilateral nonpalpable gonads is potentially a genetic female (46,XX) with congenital adrenal hyperplasia until proven otherwise. Circumcision should be avoided in the initial management. More than 70% of cryptorchid testes are palpable by physical examination and need no imaging. In the remaining 30% of cases with a nonpalpable testis, the challenge is to confirm absence or presence of the testis. At this time, there is no radiologic test that can conclude with 100% accuracy that a testis is absent.
18. **a. No change in serum testosterone following hCG stimulation.** Serum testosterone may not increase significantly in response to hCG stimulation in individuals with abnormal testes.
19. **b. History of prior testicular position provided by the PCP.** Testicular position can change over time and/or be difficult to ascertain in boys with retractile testes; therefore, a careful examination is necessary to differentiate between retractile and undescended testes.
20. **e. The majority of testes that descend spontaneously do so in the first few months of life.** For full-term boys of normal weight, spontaneous testicular descent typically occurs in the first months after birth and is rare after 6 months of age.
21. **a. 2 to 5 times the risk in normal boys.** Surgery may reduce but does not eliminate the risk of TGCT in boys with cryptorchidism and the risk exists in the contralateral testis as well, albeit lower. PLAP+ germ cells can be found in normal testes after birth.
22. c. **Mature teratoma.** The risk of benign testicular tumors is not increased in cryptorchidism.
23. **d. Perianal.** All the other answers reflect possible positions for ectopic testes that are possible although rare.

24. **c. AMH.** AMH levels remain high during childhood and are downregulated at puberty.
25. **c. 20 to 28 weeks gestation.** The testis passes into the inguinal canal at 20 to 28 weeks gestation during the fifth phase of testicular descent.
26. **b. 1% to 4%.** Cryptorchidism is one of the most common congenital anomalies occurring in 1% to 4% of full-term male infants.
27. **e. Observation until 6 months of age.** Descent may occur up until 6 months of age. If it hasn't occurred by this time, surgical treatment should be performed. Diagnostic imaging has not been shown to change the need for surgery. Hormonal therapy to induce descent is no longer supported given the lack of evidence that exists.

Chapter Review

1. Gonadal determination involves separate genetic pathways for development of testis and ovary and *SRY* is a master switch in males that regulates downstream testis-determining genes.
2. Differentiation of gonocytes, Sertoli and Leydig cells occurs between 5 and 9 weeks' gestation and the gubernaculum, the guide for testicular descent, appears at 7 weeks.
3. Levels of the Leydig cell hormones testosterone and INSL3 peak between 14 and 17 weeks, and are critical for testicular descent.
4. Swelling of the gubernaculum, which starts in the second trimester, provides space for passage of the testis into the scrotum between 20 and 28 weeks.
5. Cryptorchidism occurs in 1%–4% of full-term males; both spontaneous descent (in the first few months of life, usually by 6 months) and reascent of testes may occur.
6. The causes of cryptorchidism are largely unknown, but birth weight, gestational age, and genetic and environmental risk factors likely contribute to disease risk.

7. Acquired cryptorchidism is not uncommon but is likely equivalent to cases diagnosed at birth. It may occur after scrotal testes are confirmed at birth, or after spontaneous descent of a cryptorchid testis, and may be more common in boys with retractile testes. Yearly testicular exams are recommended during childhood.
8. About 80% of undescended testes are palpable and 60%–70% are unilateral.
9. Epididymal anomalies and a patent processus vaginalis are common in cases of cryptorchidism.
10. Subsets of boys with cryptorchidism have subtle abnormalities of the hypothalamic-pituitary-gonadal axis during mini-puberty and/or reduced levels of hormones secreted by Sertoli cells, AMH, or inhibin B (INHB).
11. Orchidopexy is recommended for testes that remain undescended after 6 months of age; hormone therapy is not recommended.
12. Laparoscopy is the procedure of choice and imaging studies are rarely useful in the diagnosis and treatment of nonpalpable abdominal cryptorchidism.
13. Immediate urologic consultation is required for all phenotypic male newborns with bilateral, nonpalpable testes for evaluation of a possible DSD. A newborn with a male phallus and bilateral nonpalpable gonads is potentially a genetic female (46,XX) with congenital adrenal hyperplasia until proven otherwise.
14. Sperm counts are reduced in at least 25% of formerly unilateral and the majority of formerly bilateral cryptorchid men, but paternity rates in the unilateral group are similar to control men.
15. Ad spermatogonia counts may predict fertility potential in some males with cryptorchidism**.**
16. TGCT risk is 2–5 times higher in boys with cryptorchidism and is positively correlated with age at surgery.